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PHISHING EMAIL DETECTION USSING LOGICAL REGRESSION

24CY731 M.TECH CYBER SECURITY SEMESTER: II

Mini Project Report Submitted by,

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ABSTRACT

The conventional defense mechanism against phishing attacks has been through email filtering techniques, where rule-based filters and signature-based mechanisms have been used. However, these techniques have become ineffective as attackers continue to compromise users' email account details.

The goal of this project is to build an Automated Phishing Detection System for Emails using a hybrid model of logistic regression machine learning technique alongside the tf-idf algorithm to extract important features from emails. We gathered a total of 13,565 emails from four different public domains including; Ling-Spam Corpus; Nazario Phishing Corpus; Nigerian Fraud Email Dataset; and SpamAssassin Corpus. From each of the emails collected, we took into consideration the subject line as well as the email body. An email cleaning process involved; converting all email contents into lower case, removing any URL link and tokens that correspond to email address, substituting any numerical figures with placeholders, removing punctuations and finally using rule-based manual lemmatization technique. Once the cleaning process was completed, the tf-idf algorithm was used to convert the email contents into vector form to produce unigram and bigram of words, resulting to a highly dimensional space of features. This gave us a 15,000-dimensional feature space. The logistic regression model we built was able to achieve; an accuracy rate of 97.83%; 96.54% recall rate in the phishing category; 99% precision rate; and finally an F1 score of 97.89%. The output probabilities of our model were then used in building a three-tier risk assessment system that can provide meaningful real-time analysis of suspicious emails. Five fold cross-validation was done giving us an average accuracy of 96.27%.

Keywords: *Phishing Detection, Logistic Regression, TF-IDF, Email Security, Machine Learning, Natural Language Processing, Risk Scoring.*

1. INTRODUCTION

These days crime is happening all the time. People are finding new ways to commit crimes, like cyber crime. One type of cyber crime is phishing. Phishing is when someone sends electronic messages to trick people into doing something bad like clicking on a bad link downloading bad software or giving away secret information. According to people who work in cybersecurity phishing is one of the common and expensive types of cyber crime and it is responsible for a lot of data breaches and ransomware attacks every year.
Phishing attacks people, not computers. The bad people pretend to be someone you trust like a bank, a technology company, a government agency or someone you work with. They try to make you think something is urgent or important. This can cause problems for individuals and big organizations like hospitals, governments and important infrastructure.
Artificial Intelligence and Machine Learning are tools for stopping cyber crime. They can look at data. Find patterns that other systems cannot. For example when it comes to finding phishing emails old machine learning models that use Natural Language Processing can do a job. They can understand what phishing emails look like. What words they use.
Even though people have done a lot of research on phishing many of the methods are not perfect. Some methods use systems that require a lot of computer power while others need people to manually create and update lists of bad words and phrases. The reason for this project is to create a phishing detection system that's accurate easy to understand and does not require a lot of computer power. It should not just say yes or no,. Also give a score that shows how likely it is that an email is a phishing attempt.
The main goals of this project are: to collect a dataset of phishing emails from different sources; to create a good system, for preparing and analyzing the text of emails; to train a Logistic Regression classifier to detect phishing emails; to create a system that gives a score showing how likely an email is to be a phishing attempt; and to test the system to make sure it works well with different types of emails.

2. LITERATURE SURVEY

A survey of existing research on email phishing detection shows studies. These studies use methods like rule-based filtering, classical machine learning, deep learning and hybrid approaches. The works reviewed below cover a range of techniques. They provide background for understanding the motivation and design behind the proposed framework.

1. Sahami, M., Dumais, S., Heckerman, D., & Horvitz E. (1998). "A Bayesian Approach to Filtering Junk E-Mail." AAAI Workshop on Learning for Text Categorization.
Sahami et al. Presented one of the works on email classification. They used Bayesian methods to filter spam. The system used a simple word representation. Learned prior probabilities of spam and legitimate word occurrences. It produced a classification result. The approach was fast and effective for simple spam filtering. However it assumes words are independent. It ignores word order and context. It can be manipulated by crafted inputs. Despite these limitations the framework established by this work underpins research in text-based threat detection.

2. Drucker, H., Wu, D., & Vapnik, V. N. (1999). "Support Vector Machines for Spam Categorization." IEEE Transactions on Neural Networks, 10(5) 1048–1054.
Drucker et al. Applied Support Vector Machines (SVMs) to email classification. They showed SVMs perform better than Bayes on benchmark datasets. SVMs identify a separating line in a high-dimensional feature space. They are robust to the curse of dimensionality of text data. The work demonstrated that margin-maximising classifiers could exploit the high-dimensional nature of word-count features. However SVMs have computational cost during training on large corpora. They are sensitive to the choice of kernel and regularisation parameters. They do not provide calibrated probability output.

3. Fette, I., Sadeh, N., & Tomasic A. (2007). "Learning to Detect Phishing Emails." Proceedings of the International Conference on World Wide Web (WWW) 649–656.
Fette et al. Proposed PILFER. It is a phishing detection system. PILFER extracts features from email headers, embedded URLs and HTML content. It classifies using a Random Forest. The system achieved precision. It leveraged signals that do not depend solely on message text. The primary limitation is reliance on URL and HTML features. It is vulnerable to text- phishing. It requires feature engineering effort per attack vector.

4. Almeida, T. A., Gomez Hidalgo, J. M., & Yamakami A. (2011). "Contributions to the Study of SMS Spam Filtering." Proceedings of the 2011 ACM Symposium on Document Engineering, 259–262.
Almeida et al. Evaluated classical classifiers on the SMS Spam Collection dataset. They used TF-IDF features. Logistic Regression emerged as one of the performing models. It produces calibrated probabilities. It handles dimensional sparse feature spaces efficiently. The study provides empirical justification for the approach adopted in the present project. However the study focuses on SMS messages. It does not cover -paragraph email bodies. It does not have a risk-scoring mechanism, beyond classification.

5. Bahnsen, A. C., Bohorquez, E. C., Villegas, S., Vargas, J., & Gonzalez, F. A. (2018). "Classifying Phishing URLs Using Recurrent Neural Networks." Proceedings of the APWG Symposium on Electronic Crime Research (eCrime).
Bahnsen and his team of researchers used Long Short-Term Memory networks to look at the order of characters in phishing emails and URLs. They found this model was really good at finding patterns that other methods could not find. It was successful in finding phishing URLs. However the model was slow. Required a lot of information to do its job.
The team also had a time figuring out how the model made its choices.
This study, by Bahnsen and his team was significant. It showed that analyzing character sequences can be a way to identify phishing attempts. The study also showed that this method can be hard to use in places where resourcesre limited. Bahnsen and his team learned that using Long Short-Term Memory networks to look at phishing emails and URLs can be helpful. It has its own set of problems.

6. Vazhayil, A., Vinayakumar, R., & Soman, K. P. (2019). "Comparative Study of the Detection of Malicious URLs Using Shallow and Deep Networks." Proceedings of the International Conference on Smart Computing and Control Systems.
Vazhayil et al. and others have looked into using BERT for transfer learning to find phishing. BERT models are trained on huge amounts of text and can capture rich contextual representations. They always do very well on text classification benchmarks. The research validated that contextual embeddings significantly surpass shallow feature techniques in URL and content classification. But BERT-class models need GPU acceleration, have hundreds of millions of parameters, and are hard to understand, which makes it hard to use them in lightweight or regulated settings.

7. Toolan, F., & Carthy, J. (2010). "Feature Selection for Spam and Phishing Detection." 5th Conference on Email and Anti-Spam (CEAS).
Toolan and Carthy investigated ensemble methods including Random Forests and boosted decision trees for phishing email classification, alongside systematic feature selection analysis. Ensembles achieved high accuracy by reducing variance through aggregation. The study found that a relatively small number of well-chosen features could achieve performance competitive with full feature sets, with implications for computational efficiency. Limitations include increased computational cost relative to single classifiers and reduced model interpretability compared to linear approaches.

8. Abu-Nimeh, S., Nappa, D., Wang, X., & Nair, S. (2007). "A Comparison of Machine Learning Techniques for Phishing Detection." Proceedings of the Anti-Phishing Working Groups 2nd Annual eCrime Researchers Summit, 60–69.
Abu-Nimeh et al. conducted a comparative study of six machine learning algorithms using a hybrid feature set combining textual, URL-structural, and header-based features. The study found that models combining multiple feature modalities outperformed those relying on text alone, with Logistic Regression and Random Forests performing competitively across metrics. The primary limitation is significant engineering overhead in curating and maintaining multi- modal extraction pipelines that must be updated as email formats and attack vectors evolve.

9. Alam, M. N., Sarma, D., Islam, M. F., Saha, I., Ullah, R., & Hossain, M. I. (2020). "Phishing Attacks Detection – A Machine Learning-Based Approach." 2020 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE).
Alam et al. applied Convolutional Neural Networks to phishing detection by treating word embedding sequences as one-dimensional signals and applying convolutional filters to extract local n-gram patterns. CNNs achieve competitive accuracy and are faster to train than LSTMs. The work demonstrated that local pattern detection via convolution is effective for capturing characteristic phishing phrases. Limitations include the absence of sequential modelling capability and substantially higher resource requirements than classical approaches such as Logistic Regression with TF-IDF.

10. Bergholz, A., De Beer, J., Glahn, S., Moens, M. F., Paass, G., & Strobel, S. (2010). "New Filtering Approaches for Phishing Email." Journal of Computer Security, 18(1), 7–35.
Bergholz et al. proposed a topic modelling approach using Latent Dirichlet Allocation (LDA) to discover latent phishing topics in email corpora, using these topic distributions as features for a downstream classifier. The unsupervised topic discovery adds robustness against novel phishing campaigns not represented in the training set. Limitations include sensitivity to the number of topics as a hyperparameter and the need for large datasets to produce stable topic distributions. The study contributes an important unsupervised dimension to the phishing detection literature.

2.1 Research Gap

Most of the literature reviewed revolves around binary classification of whether an incoming message is either phishing or legitimate without much understanding of the severity of the attack. In addition to this, many systems built upon URL and header properties have complex multimodal feature extraction pipelines which incur high maintenance cost. Moreover, many of the deep learning based systems require high computational power making it infeasible to deploy them in email server environment with restricted resources. Finally, most systems developed so far do not generate probabilistic risk scores but generate binary outcomes. This research gap can be addressed by developing an effective email phishing detection system which not only produces accurate classification but also generates meaningful risk score.

3. PROBLEM STATEMENT

The phishing email attack vector has recently escalated as a significant security problem, whereby attackers use deception via email to trick victims into sharing credentials, financial information, and personal data. While considerable effort has been invested in researching and implementing commercial anti-phishing filters in recent decades, such systems still fail to prevent phishing attacks. This is mainly because they are unable to adapt to new phishing attack vectors outside their training distribution without constant manual oversight.

Existing solutions for detecting phishing emails tend to be limited in their classification capability to determining whether the message is legitimate or not. They offer no indication of the type of attack or its confidence level. Their inability to provide a calibrated probability score makes it difficult for the user to assess different attacks' severity. In addition, many accurate models are computationally expensive and impractical to deploy on the hardware-limited mail server infrastructure. Moreover, multi-modal feature extractors like URL parsers and HTML tree structures cannot process non-structured phishing texts.

Therefore, there is a need for an effective and lightweight phishing detector system that not only accurately detects phishing emails based on text content alone but also generates a probability-based risk score allowing prioritised threat response.

4. PROPOSED METHODOLOGY

In the proposed method, a phishing email detection system is proposed that combines the traditional Natural Language Processing methods with Logistic Regression as the classifier and a three-stage risk assessment process based on probabilities. The main aim behind the implementation of this model is to detect phishing emails along with providing a confidence score regarding the severity of the detected attack, in a computationally efficient manner without requiring any special hardware setup.

4.1 System Architecture

As shown in Figure 4.1, the architecture of the proposed model consists of the following stages. First of all, raw data consisting of email data are loaded from four different sources and prepared in one dataset. Text data are then preprocessed and extracted using TF-IDF Vectorisation. After preparing 15,000 dimensions of sparse features, these features are provided to the Logistic Regression classifier, which predicts the phishing score.

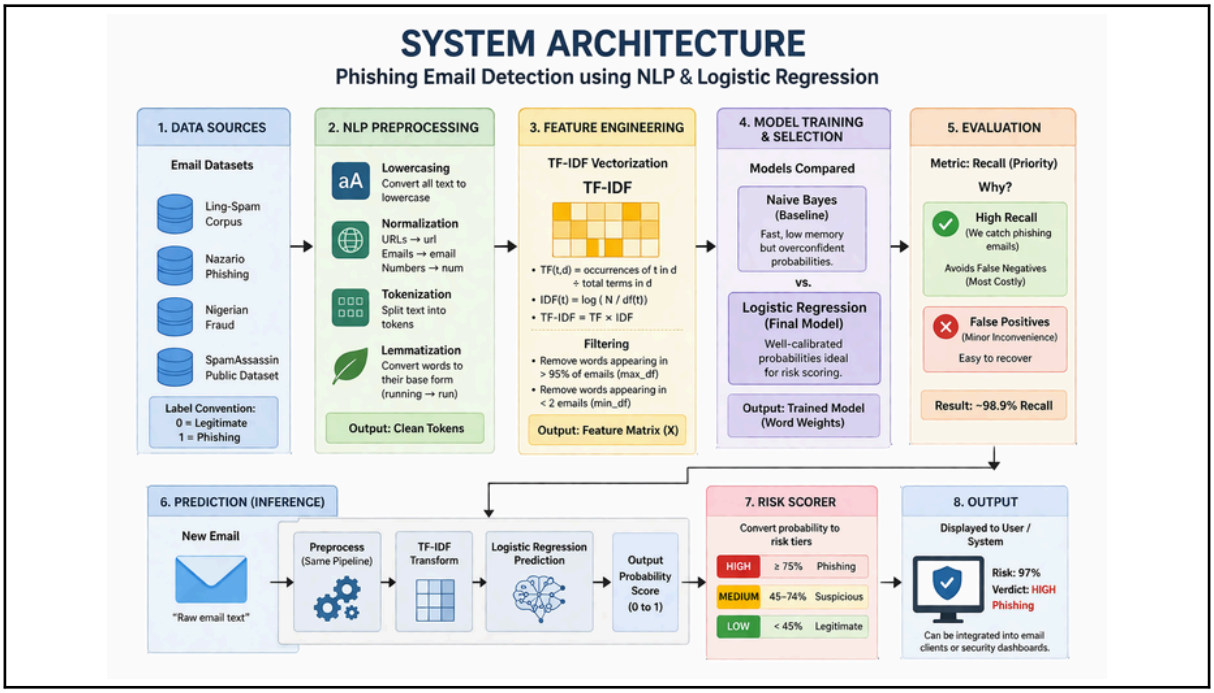

Fig 4.1 Overall Pipeline of the Phishing Email Detection System

4.2 Dataset

The dataset was constructed by combining four publicly available sources from the Kaggle Phishing Email Dataset collection (Naser Abdullah Alam, 2023). The total number of email messages is 13,565. The combined dataset encompasses two primary classes: Legitimate (label 0) and Phishing (label 1). Each message is represented using the concatenated subject line and body text of the email.

Dataset	Total Emails	Legitimate (0)	Phishing (1)	Category
Ling-Spam	2,859	2,401	458	Spam / Legit
Nazario	1,565	0	1,565	Phishing
Nigerian Fraud	3,332	0	3,332	Fraud
SpamAssassin	5,809	4,091	1,718	Spam / Legit
Total	13,565	6,492	7,073	—

4.3 Data Preprocessing

Before training, the raw emails dataset passed through several preprocessing stages. The email's subject and body text fields were concatenated into one. In the cleaning pipeline, the following stages have been applied sequentially. Lowercasing all letters to unify different cases. URL, email addresses, and numbers have been replaced with tokens 'url', 'email', and 'num'. All punctuation marks have been removed. Rule-based Lemmatization of English words was done by removing the most common morphological endings such as '-ing', '-tion', '-ness', '-ment', '-ed', '-er', and '-ly'. Tokens shorter than three characters have been removed as they don't carry any semantic value when it comes to phishing.

4.4 Feature Extraction: TF-IDF Vectorisation

For each preprocessed email, a numerical representation was obtained through Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. This algorithm assigns each term a score based on its importance calculated relative to the whole document collection. So, it gives low values to uninformative terms and higher values to more important ones. Here are the parameters used in this vectorization method: Vocabulary size limit set to 15,000. N-gram range equal to (1,2), meaning that we include not only individual words but pairs of consecutive words as well like click here or verify account. Sublinear TF scaling is used to avoid giving proportionally high values to very frequent terms. Minimum document frequency threshold equals 2. Maximum document frequency threshold equals 0.95. These parameters have resulted in the creation of training matrix with shape 10,852 x 15,000.

4.5 Data Splitting

The combined dataset was divided into two subsets. The training set comprised 80% of the total data (10,852 emails), which was used for fitting the TF-IDF vectoriser and training the classifier. The test set consisted of the remaining 20% (2,713 emails), held out entirely for final evaluation. A stratified split was applied so that the class distribution of legitimate and phishing emails was preserved across both subsets.

4.6 Model Training: Logistic Regression

A Logistic Regression with parameter C=0.1 was used to train our classification model. Low value of parameter C leads to larger regularization which means that our model will generate more moderate and dispersed probability scores. This makes our classification model more suitable for risk scoring purposes. Solver was limited to 1,000 iterations to ensure it converges. The model learns weights for each of 15,000 TF-IDF terms and outputs phishing probability calculated with the help of logistic sigmoid function.

4.7 Risk Scoring System

The continuous phishing probability generated by the Logistic Regression model was converted into a risk score based on a structured three-tiered scoring process. Emails having a predicted probability of phishing equal to or greater than 75% are categorised as HIGH risk and marked as Phishing. Emails with a probability from 45% up to 74% are categorised as MEDIUM risk and also marked as Phishing. On the other hand, emails with a probability less than 45% are categorised as LOW risk and marked as Legitimate.

5. RESULTS AND DISCUSSION

The model demonstrated strong performance on the held-out test set of 2,713 emails, achieving an overall accuracy of 97.83%. The training and evaluation results are presented in detail across the following subsections.

5.1 Class Distribution and Dataset Overview

The combined dataset of 13,565 emails is slightly imbalanced, with 7,073 phishing samples (52.1%) and 6,492 legitimate samples (47.9%). Figure 5.1 presents the class distribution bar chart and per-dataset composition. The near-balanced split ensures that the classifier is not biased toward the majority class during training.

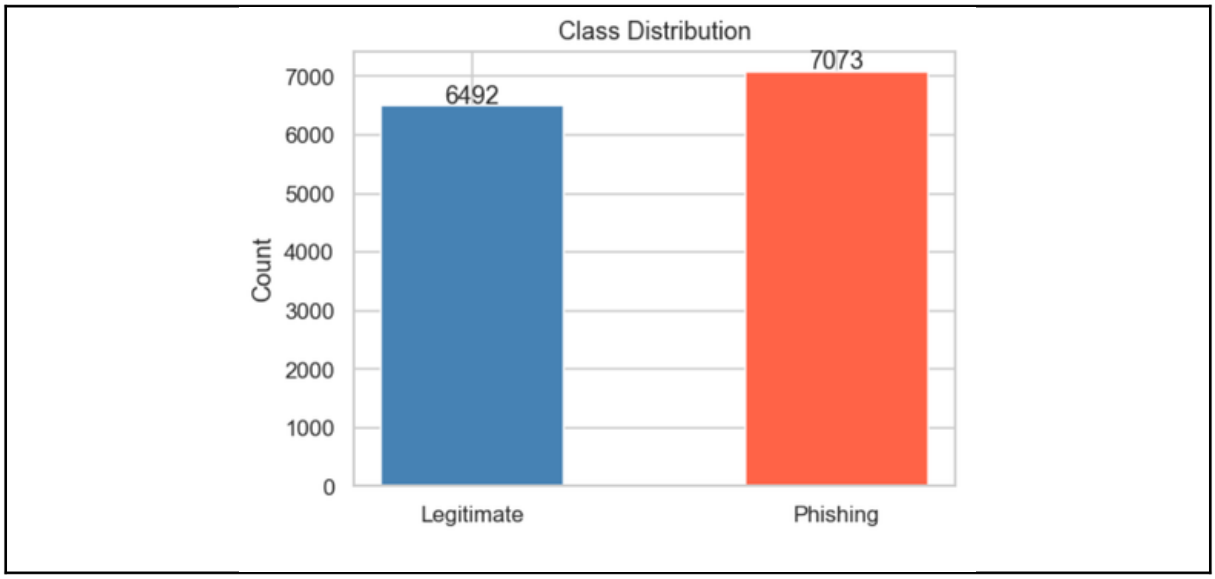

Fig 5.1 Class Distribution and Dataset Overview

5.2 Email Length Distribution

The email length distribution is shown in Figure 5.2 for legitimate emails and phishing emails, with emails that exceed 5,000 characters removed. As expected, phishing emails are shorter, often containing only one or two sentences. The length distribution is wider for legitimate emails. This fact suggests using bigram features to detect typical phrases used in phishings emails.

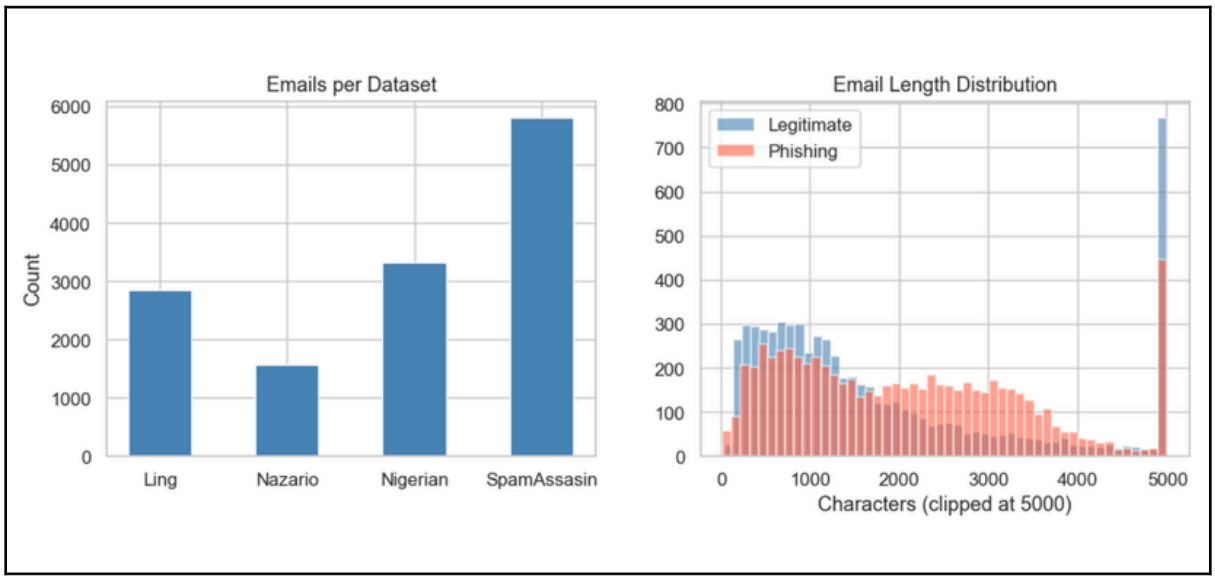

Fig 5.2 Email Distribution by Class

5.3 Test Set Classification Performance

The Logistic Regression model was evaluated on the held-out test set. The following table presents the full classification report including per-class precision, recall, and F1-score. Consistent with the training design, recall for the phishing class is prioritised as the primary metric, since a false negative (an undetected phishing email) causes greater harm than a false positive.

Class	Precision	Recall	F1-Score	Support
Legitimate	0.96	0.99	0.98 0.98	1,298
Phishing	0.99	0.97	0.98	1,415
Macro Avg	0.98	0.98		2,713
Overall Accuracy: 97.83%				**2,713**

The model correctly classified 2,654 of 2,713 test emails. The phishing class achieved a precision of 99%, meaning that when the model predicted phishing it was correct 99% of the time, while the recall of 96.54% indicates that approximately 97 out of every 100 genuine phishing emails were correctly identified. The confusion matrices from the evaluation are presented in Figures 5.3.

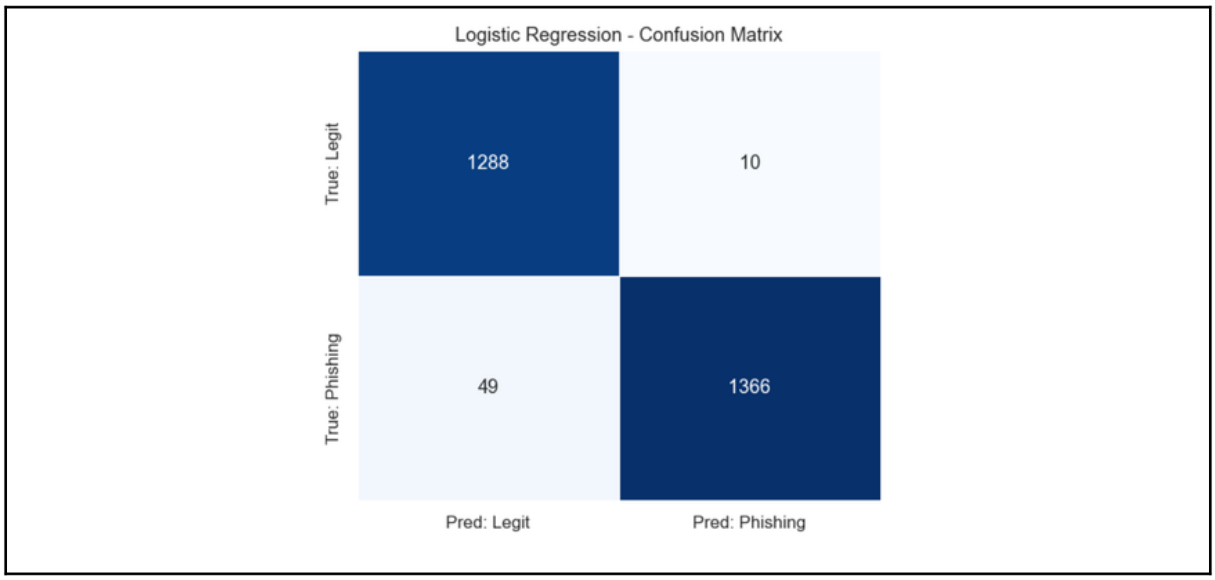

Fig 5.3 Confusion Matrix – Logistic Regression

5.4 ROC Curve and Probability Distribution

Figure 5.4 presents the Receiver Operating Characteristic (ROC) curve for the Logistic Regression model. The Area Under the Curve (AUC) reflects the model's ability to discriminate between legitimate and phishing emails across all classification thresholds. Figure 5.5 shows the distribution of predicted phishing probabilities for both classes, confirming that the model produces well-separated probability distributions, supporting the validity of the three-tier risk scoring thresholds.

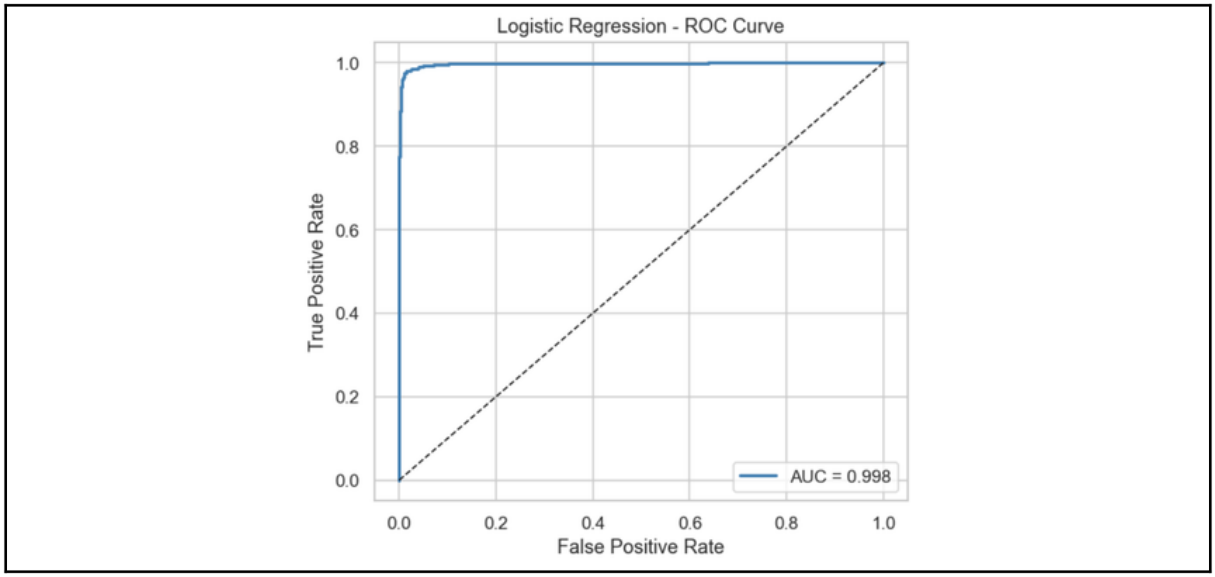

Fig 5.4 ROC Curve – Logistic Regression

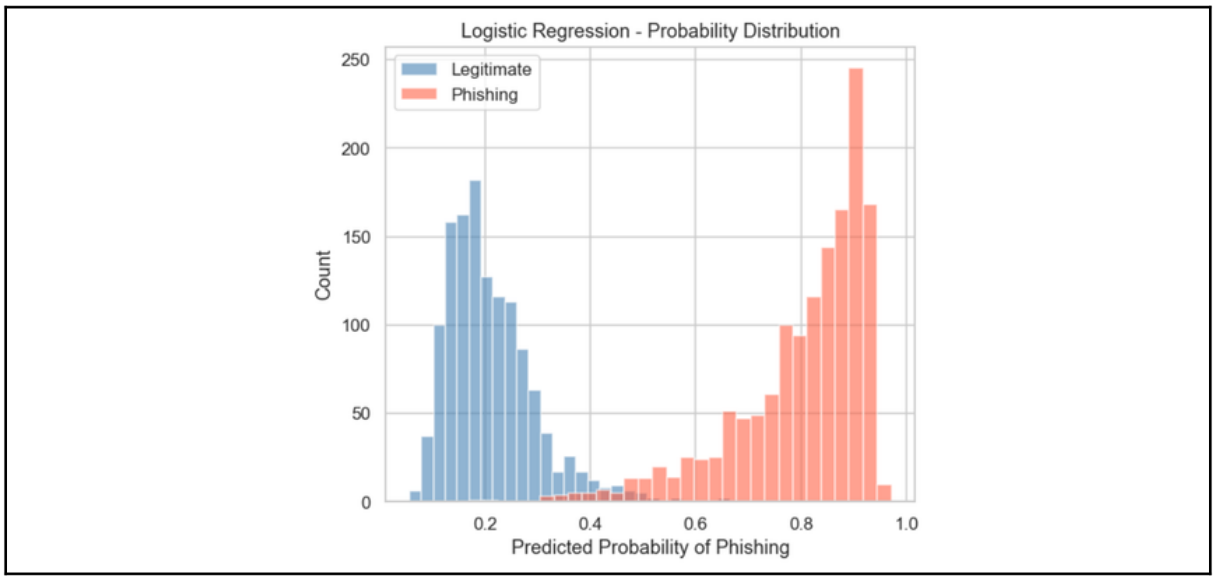

Fig 5.5 Probability Distribution of Predicted Phishing Scores

5.5 Cross-Validation Results

To validate the generalisability of the model beyond the single train-test split, five-fold cross-validation was performed on the entire dataset using a scikit-learn Pipeline combining the TF-IDF vectoriser and Logistic Regression classifier. This ensures that the vectoriser is fitted only on training folds during each iteration, preventing data leakage. Figure 5.6 shows the training history across folds.

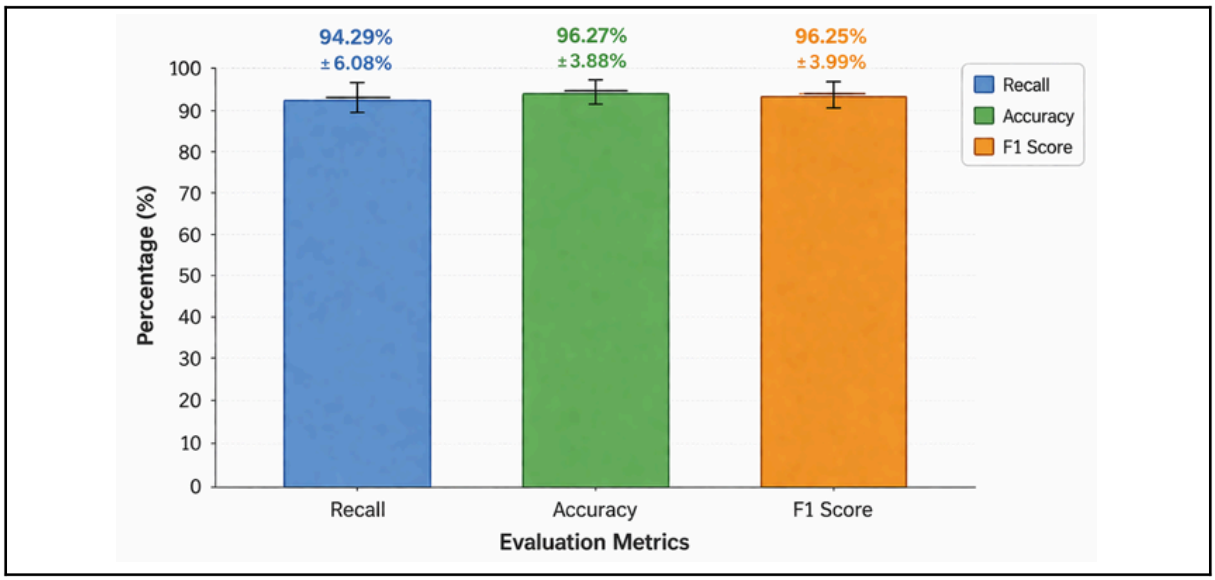

Fig 5.6 5-Fold Cross-Validation Training History

Metric	Mean	Std Dev
Accuracy	96.27%	+/- 3.88%
Recall	94.29%	+/- 6.08%
F1-Score	96.25%	+/- 3.99%

The cross-validation results confirm that the model generalises well across different data splits. The mean accuracy of 96.27% across five folds closely tracks the single-split result of 97.83%, indicating no overfitting to any particular data partition.

5.6 Risk Scorer Analysis

The three-tier risk scoring system was validated on six representative test cases. The results demonstrate the practical utility of the probability output for graduated threat assessment.

Email Type	Sample Text (truncated)	Score	Level	Verdict
Credential Phishing	Your account has been suspended...	91.9%	HIGH	Phishing
Nigerian Fraud	Dear Beneficiary, You have inherited...	76.3%	HIGH	Phishing
Security Notice	We noticed a login from a new device...	76.3%	HIGH	Phishing
Delivery Notice	Your package could not be delivered...	67.2%	MED	Phishing
Legitimate (Meeting)	Hi Sarah, the meeting is at 3 PM...	36.6%	LOW	Legitimate
Legitimate (Report)*	The quarterly report has been attached...	59.1%	MED	Phishing*

- The quarterly report email was classified as phishing at MEDIUM risk (59.1%). This illustrates a known limitation of pure text-based TF-IDF models: financial language in legitimate emails co-occurs with phishing vocabulary in the training corpus. This false positive would be addressed in future work through sender reputation or contextual features.

6. CONCLUSION AND FUTURE WORK

The proposed system presents a lightweight and efficient phishing email detection approach using a TF-IDF feature extraction pipeline combined with a Logistic Regression classifier. The model successfully classifies emails as legitimate or phishing with an accuracy of 97.83%, a phishing class precision of 99%, and a recall of 96.54%. By making use of probabilistic output, the system achieves effective risk scoring while maintaining computational efficiency and interpretability. The three-tier risk assessment design: High, Medium, and Low, improves both accuracy and practicality in comparison to binary classification methods that provide no confidence information. Five-fold cross-validation yielded a mean accuracy of 96.27%, confirming the robustness of the approach. Overall, the proposed methodology facilitates a more informative and deployable solution for phishing email detection using text-only features, without requiring GPU infrastructure or complex multi-modal feature engineering.

While the proposed system demonstrates effective phishing detection, there remains substantial scope for further enhancement and extension. The model can be improved to incorporate multi-modal features combining textual TF-IDF scores with URL reputation signals, sender domain age, and DMARC/SPF authentication metadata, which would substantially reduce false positives on emails containing financial language. Experimentation with advanced NLP models such as DistilBERT or fine-tuned BERT variants could be explored on the same dataset to quantify the accuracy-interpretability trade-off more precisely. The development of an online learning component that incrementally updates model weights as new confirmed phishing emails are identified would improve robustness against zero-day campaigns. Real-time deployment integration with email server platforms such as Postfix or Microsoft Exchange through a REST API wrapper would enable practical evaluation of the system's throughput and operational false-positive rate in production environments. Additionally, extending the dataset to include multilingual phishing emails would broaden the system's applicability to non-English speaking organisations, and integration with a centralised threat reporting system would allow automatic aggregation of detected phishing campaigns.

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